Class 12 Chemistry

NCERT syllabus 2026-27 par based

Chemophile

Part 2 (for 12th Class)

FREE SAMPLE
har chapter se ek topic

Har chapter ki ek jhalak

NCERT Class 12 Chemistry ke saare 10 chapters, easy Hinglish mein, har page par visuals ke saath



1 Solutions
2 Electrochemistry
3 Chemical Kinetics
4 The d- and f-Block Elements
5 Coordination Compounds
6 Haloalkanes and Haloarenes
7 Alcohols, Phenols and Ethers
8 Aldehydes, Ketones and Carboxylic Acids
9 Amines
10 Biomolecules

575 colourful pages
313 figures aur diagrams
298 solved examples
990+ practice questions
10 board pattern tests (40 marks)

Adeel Sir

Mohd Adeel Khan
MSc BEd (JMI) | Ph. **9953821875**

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Chemophile kya hai?

Chemophile Part 2 (for 12th Class) NCERT Class 12 Chemistry ke saare **10 chapters** ki colourful booklets hain. Explanations **modern Hinglish** mein, technical terms aur exam answers **English** mein. Har topic ek story ki tarah chalta hai: Adeel Sir ka example, visual, concept, exam wala answer, aur end mein practice.

10	575	313	298	990+
chapters	colourful pages	figures aur diagrams	solved examples	practice questions

Har page par kya milega

Adeel Sir ka example	Har concept ek everyday life example se start hota hai.
⚠ Galti mat karna	Common mistakes jo sabse zyada marks cut karwati hain.
Adeel Sir ka solved example	Step-by-step solved question.
Ruko aur socho	Aage padhne se pehle khud socho, phir tick karo.
Exam mein aise likho	Board exam ke liye English answer, ruled lines par.
★ JEE/NEET extra	Advanced concepts, competitive exams ke liye.
Adeel Sir ka tip	Yaad rakhne ki short trick.
Ab tumhari baari	Khud solve karo; answers topic ke end mein.

Har chapter ke andar

- ✓ Har topic ka pehla page: goals, timeline aur Adeel Sir ka message
- ✓ NCERT ke saare Examples, Intext Questions aur Exercises solved
- ✓ JEE/NEET extra boxes aur common mistakes
- ✓ Har chapter ke end mein revision: fact ya formula sheet, top mistakes
- ✓ Har concept: everyday example, phir visual, phir concept
- ✓ Exam mein aise likho: board ke liye English answers
- ✓ Har topic ke end mein recap, quick check aur answers
- ✓ 40 marks ka board pattern test, answer key aur certificate

Is sample mein kya hai?
Har chapter se **ek topic** ke do asli pages: topic ka pehla page aur uska pehla lesson page. Phir Chapter 5 ke revision ke do pages (fact sheet aur board pattern test ka pehla page). Yeh pages bilkul waise hi hain jaise poori book mein: **575 pages** isi style mein.

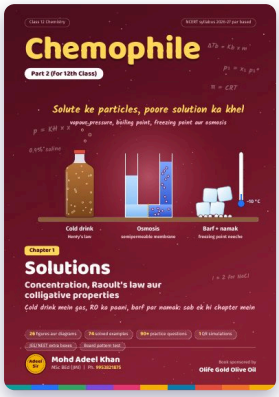
Adeel Sir kehte hain
Chemistry ratne ka subject nahi, samajhne ka subject hai. Jab concept dikh jaata hai, to formula khud yaad ho jaata hai.

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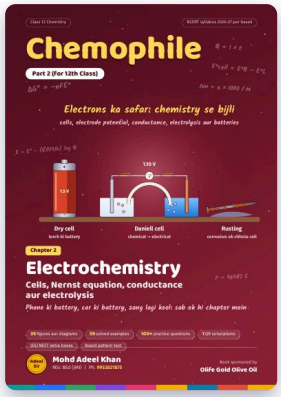
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Saare 10 chapters ek nazar mein

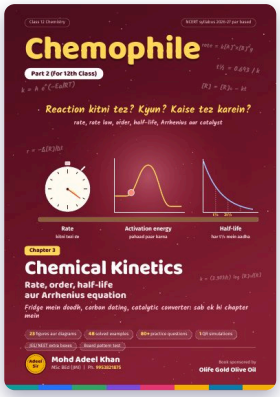
Har chapter ki apni booklet hai. Neeche har booklet ka cover, uske pages aur is sample mein shaamil topic.



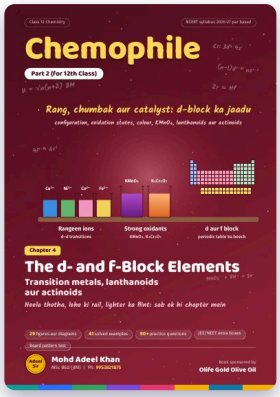
Chapter 1 Solutions
66 pages · Sample: Osmosis aur osmotic pressure



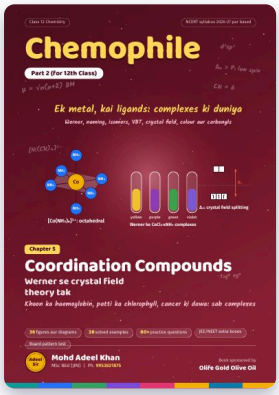
Chapter 2 Electrochemistry
59 pages · Sample: Nernst equation



Chapter 3 Chemical Kinetics
53 pages · Sample: First order reactions aur half-life



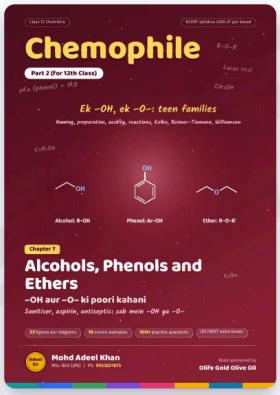
Chapter 4 The d- and f-Block Elements
58 pages · Sample: Potassium dichromate aur permanganate



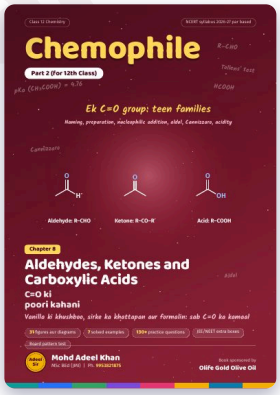
Chapter 5 Coordination Compounds
55 pages · Sample: Crystal field theory



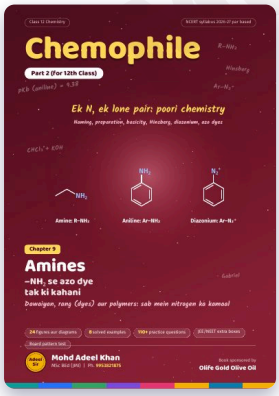
Chapter 6 Haloalkanes and Haloarenes
60 pages · Sample: Nucleophilic substitution: SN2 aur SN1



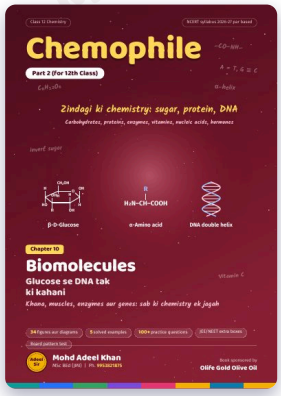
Chapter 7 Alcohols, Phenols and Ethers
55 pages · Sample: Acidity aur O–H bond ki reactions



Chapter 8 Aldehydes, Ketones and Carboxylic Acids
64 pages · Sample: α-Hydrogen, Cannizzaro aur uses



Chapter 9 Amines
52 pages · Sample: Diazonium salts



Chapter 10 Biomolecules
53 pages · Sample: Nucleic acids

Poori series

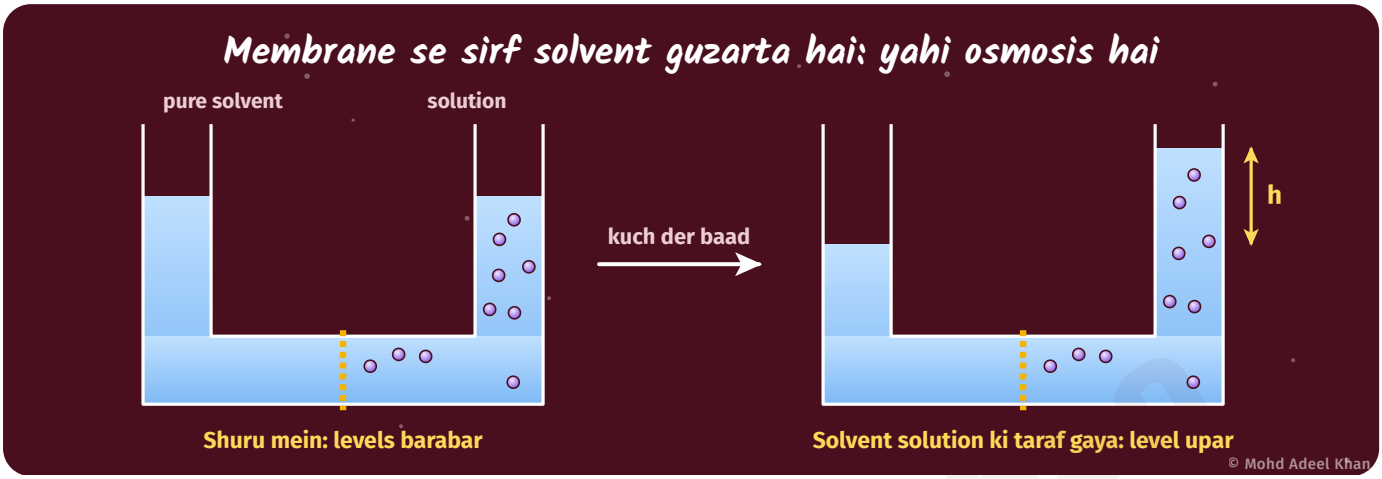
10 chapters, 575 pages, har chapter ek alag booklet. Har booklet mein cover, contents, topics, revision aur board pattern test.

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Topic 8

Osmosis aur osmotic pressure

Ek patli si membrane, aur paani ka safar



Kuch membranes (jaise pig's bladder, parchment, cellophane) sirf **solvent** ko guzarne deti hain, solute ko nahi: inhe **semipermeable membrane (SPM)** kehte hain. SPM ke ek taraf pure solvent aur doosri taraf solution ho, to solvent solution ki taraf behta hai: yahi **osmosis** hai. Is flow ko rokne ke liye jo extra pressure chahiye, woh **osmotic pressure (π)** hai. Is topic mein π = CRT, isotonic solutions aur reverse osmosis.

Is topic mein

Osmosis aur π = CRT	48
Isotonic solutions aur reverse osmosis	49
Example 1.11, Intext 1.12, Exercise 1.22	50
Ek nazar mein + quick check	51

Padhne ke baad tick karo

- [ ] Osmosis aur osmotic pressure define kar sakta hoon
- [ ] π = CRT se molar mass nikaal sakta hoon (Example 1.11, Intext 1.12)
- [ ] Isotonic, hypertonic aur hypotonic solutions samjha sakta hoon
- [ ] Reverse osmosis aur desalination explain kar sakta hoon

Adeel Sir

Adeel Sir kehte hain

Paani hamesha **kam concentration se zyada concentration** ki taraf jaata hai (jahan solvent kam hai). Yeh ek line yaad rakho, sab examples samajh aa jaayenge.

Kab kya hua

1748
Nollet: pig's bladder

1887
van't Hoff: π = CRT

Aaj
RO water purifier

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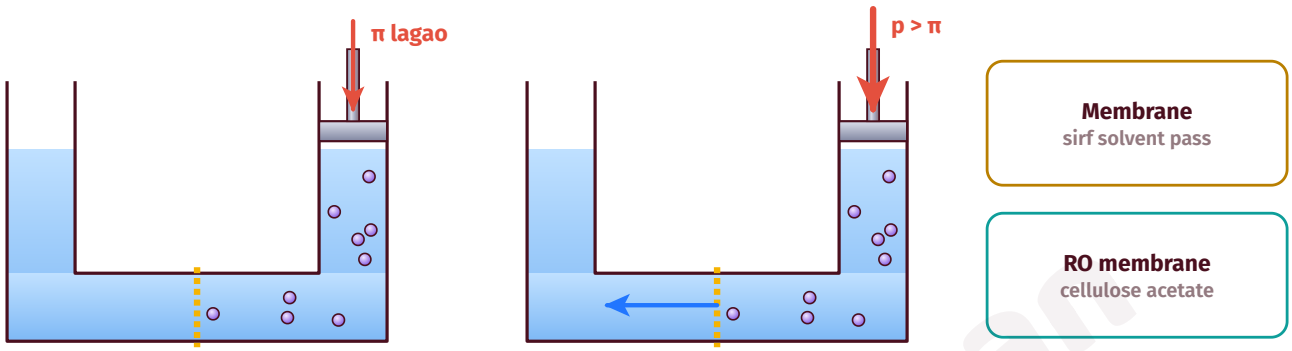
Section 1.6.4, Fig. 1.9, 1.10

Samajh aaya? Rate karo ☆ ☆ ☆

Osmosis aur osmotic pressure

Adeel Sir ka example

Adeel Sir ne raat ko **kishmish (raisins)** paani mein bhigoyi; subah woh phool kar moti ho gayi. Kishmish ki skin semipermeable jaisi hai: andar cheeni ka gaadha solution, bahar saada paani. Paani andar gaya: **osmosis**!



(a) π: solvent ka flow ruk jaata hai

(b) reverse osmosis: solvent bahar

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Osmosis

Semipermeable membrane ke through **solvent molecules** ka flow, pure solvent se solution ki taraf (ya dilute se concentrated ki taraf). Yeh tab tak chalta hai jab tak equilibrium na aa jaaye.

Osmotic pressure (π)

Woh **excess pressure** jo solution ki taraf lagaana padta hai taaki osmosis **ruk jaaye**. Yeh bhi colligative property hai: dilute solution mein yeh **molarity** ke proportional hai.

$$\pi = C R T$$

C = molarity (mol L^{-1})

$$\pi V = n_2 R T$$

$n_2 = w_2/M_2$

$$M_2 = \frac{w_2 R T}{\pi V}$$

R = 0.083 L bar mol^{-1} K^{-1}

Macromolecules ke liye osmotic pressure hi kyun?

(1) Measurement **room temperature** par hota hai, isliye proteins jaise **biomolecules** kharab nahi hote (garam karne par woh unstable hain). (2) Molality ki jagah **molarity** chalti hai. (3) Bahut dilute solution mein bhi π ka magnitude **naapne layak** hota hai, jabki ΔT_b, ΔT_f bahut chhote aate. Isliye polymers aur proteins ka molar mass isi se nikalte hain.

Exam mein aise likho

Osmosis and osmotic pressure

The flow of solvent molecules through a semipermeable membrane from pure solvent to the solution is called osmosis. The excess pressure that must be applied to the solution to prevent osmosis, i.e. to stop the passage of solvent molecules through the semipermeable membrane into the solution, is called the osmotic pressure. For dilute solutions $\pi = CRT = (n_2/V)RT$, so $M_2 = w_2RT/(\pi V)$. The osmotic pressure method is widely used to determine molar masses of proteins, polymers and other macromolecules.

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Topic 3

Nernst equation

Jab concentrations 1 M na hon

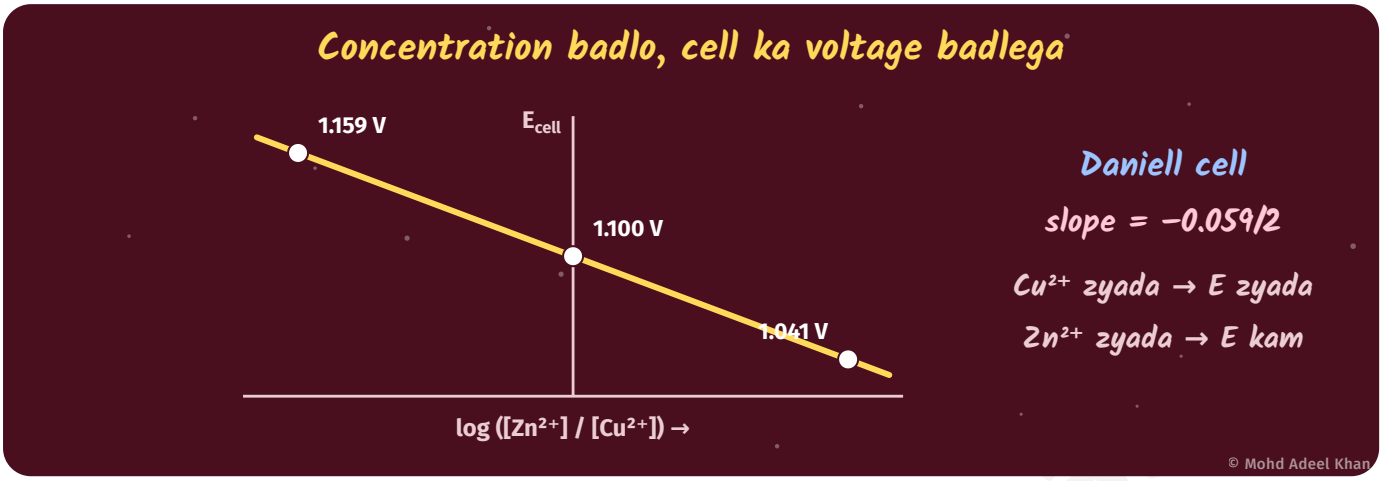


Table 2.1 ki E° values tab ki hain jab sab ions **1 M** hon. Real cells mein concentrations badalti rehti hain: battery chalte-chalte reactant ions kam hote hain aur product ions badhte hain, isliye voltage dheere-dheere girta hai. **Nernst equation** batati hai ki kisi bhi concentration par electrode potential aur cell potential kitna hoga.

Is topic mein

Nernst equation: electrode aur cell	17
Example 2.1, Intext 2.4, 2.5	18
Exercise 2.5: chaar cells ka emf	19
Ek nazar mein + quick check	20

Padhne ke baad tick karo

- ☐ Electrode aur cell ke liye Nernst equation likh sakta hoon
- ☐ Reaction quotient Q sahi likh sakta hoon (solid, liquid, gas ka dhyaan)
- ☐ Example 2.1, Intext 2.4, 2.5 jaise numericals solve kar sakta hoon
- ☐ Exercise 2.5 ke chaaron cells ka emf nikaal sakta hoon

Adeel Sir

Adeel Sir kehte hain

Nernst ka saar: **E = E° − (0.059/n) log Q** (298 K par). Q badha to E ghata. Bas Q likhna sahi aana chahiye.

Kab kya hua

1889
Walther Nernst ka equation

1920
Nernst ko Nobel Prize

Aaj
pH meter, sensors, batteries

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Section 2.3

Samajh aaya? Rate karo ☆ ☆ ☆

Nernst equation

Adeel Sir ka example

Torch ki nayi battery 1.5 V deti hai, par kuch ghante baad roshni kam ho jaati hai. Reaction to wahi hai, par andar **concentrations badal gayi**. Nernst equation isi badlaav ka hisaab hai.

Ek electrode ke liye

$M^{n+}(aq) + ne^- \rightarrow M(s)$

$$E = E^\circ - \frac{RT}{nF} \ln \frac{1}{[M^{n+}]}$$

[M] solid = 1

Kisi bhi cell ke liye

$aA + bB \rightarrow cC + dD$ (n electrons)

$$E_{cell} = E^\circ_{cell} - \frac{RT}{nF} \ln Q$$

$Q = [C]^c[D]^d / [A]^a[B]^b$

$$E_{cell} = E^\circ_{cell} - \frac{0.059}{n} \log Q$$

298 K par: 2.303RT/F = 0.059 V

$$E_{cell} = E^\circ_{cell} - \frac{0.059}{2} \log \frac{[Zn^{2+}]}{[Cu^{2+}]}$$

Daniell cell

Daniell cell ka equation kaise aaya

Cathode: $E_{Cu} = E^\circ_{Cu^{2+}|Cu} - (RT/2F) \ln (1/[Cu^{2+}])$ | Anode: $E_{Zn} = E^\circ_{Zn^{2+}|Zn} - (RT/2F) \ln (1/[Zn^{2+}])$

$E_{cell} = E_{Cu} - E_{Zn} = (E^\circ_{Cu} - E^\circ_{Zn}) - (RT/2F)[\ln (1/[Cu^{2+}]) - \ln (1/[Zn^{2+}])] = \mathbf{E^\circ_{cell} - (RT/2F) \ln ([Zn^{2+}]/[Cu^{2+}])}$

R = 8.314 J K^{-1} mol^{-1}, F = 96487 C mol^{-1}, T kelvin mein, n = transfer hone wale electrons. Daniell cell mein **$[Cu^{2+}]$ badhao** ya **$[Zn^{2+}]$ ghatao** to E_{cell} badhta hai.

Q mein kya likhein

Solid / pure liquid	Aqueous ion	Gas
[M] = 1	**molar conc.**	**partial pressure**
Zn(s), Cu(s), $Br_2(l)$, $H_2O(l)$	$[Zn^{2+}]$, $[H^+]$ in mol L^{-1}	$p(H_2)$ in bar

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⚠ **Galti mat karna.** Q mein **products upar**, reactants neeche, aur har concentration ki **power = coefficient**. $2Ag^+$ ho to $[Ag^+]^2$ likho.

Ruko aur socho

For a Daniell cell, which change increases E_{cell}?

☐ increasing $[Zn^{2+}]$ ☐ increasing $[Cu^{2+}]$

☐ making both 0.1 M

Exam mein aise likho

Nernst equation

For the electrode reaction $M^{n+}(aq) + ne^- \rightarrow M(s)$, the electrode potential at any concentration is given by $E = E^\circ - (RT/nF) \ln (1/[M^{n+}])$. For a cell reaction $aA + bB \rightarrow cC + dD$, $E_{cell} = E^\circ_{cell} - (RT/nF) \ln Q$, where $Q = [C]^c[D]^d/[A]^a[B]^b$. At 298 K, $E_{cell} = E^\circ_{cell} - (0.059/n) \log Q$.

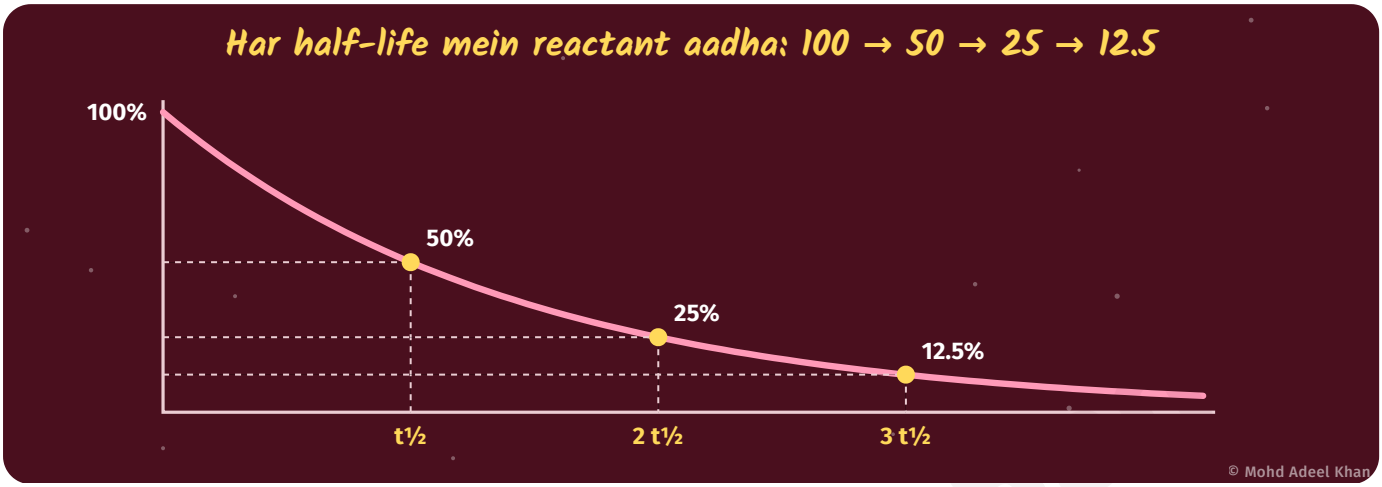
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Topic 5

First order reactions aur half-life

Har barabar time mein barabar fraction khatam



First order reaction mein rate reactant ke concentration ke **directly proportional** hai: rate = k[R]. Isliye jaise-jaise reactant ghatta hai, rate bhi ghatta hai, aur concentration **exponentially** girta hai. Is topic mein integrated equation $k = (2.303/t)\log([R]_0/[R])$, gas-phase wala pressure formula, **half-life $t_{½} = 0.693/k$**, carbon dating, aur **pseudo first order** reactions.

Is topic mein

First order: derivation, graphs, examples	**26**
First order numericals (Examples 3.5, 3.6; Intext 3.5)	**27**
Half-life (Examples 3.7, 3.8; Intext 3.6; Exercises 3.13, 3.16, 3.18)	**28**
Exercises 3.14, 3.17, 3.19, 3.24, 3.25	**29**
Exercises 3.15, 3.20, 3.21 (data aur pressure)	**30**
Pseudo first order (Example 3.9, Exercise 3.8)	**31**
Ek nazar mein + quick check	**32**

Padhne ke baad tick karo

- ☐ First order ka integrated equation derive aur use kar sakta hoon
- ☐ Gas phase reaction mein pressure se k nikaal sakta hoon (Example 3.6, Exercises 3.20, 3.21)
- ☐ $t_{½} = 0.693/k$ laga kar NCERT numericals kar sakta hoon (Exercises 3.13 to 3.19, 3.24, 3.25)
- ☐ Pseudo first order reaction samjha sakta hoon (Example 3.9, Exercise 3.8)

Adeel Sir

Adeel Sir kehte hain

First order ki pehchaan: **$t_{½}$ concentration par depend nahi karta**. 100 g ho ya 1 g, aadha hone mein same time.

Kab kya hua

1902 — Rutherford–Soddy: radioactive decay first order

1949 — Libby: carbon-14 dating

Aaj — medicine ki dose, nuclear waste

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Section 3.3.2

Samajh aaya? Rate karo ☆ ☆ ☆

First order reaction

Derivation

R → P, rate = $-d[R]/dt = k[R] \Rightarrow d[R]/[R] = -k\,dt$

Integrate: $\ln[R] = -kt + I$; $t = 0$ par $I = \ln[R]_0$

$\ln[R] = -kt + \ln[R]_0 \Rightarrow \mathbf{[R] = [R]_0\, e^{-kt}}$

$$\mathbf{k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}}$$

Do times ke beech

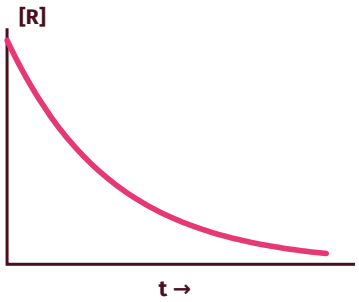
t_1 par $[R]_1$, t_2 par $[R]_2$:

$$k = \frac{2.303}{t_2 - t_1} \log \frac{[R]_1}{[R]_2}$$

Concentration ki jagah koi bhi proportional quantity (mass, moles, pressure, %) le sakte ho, kyunki **ratio** aata hai.

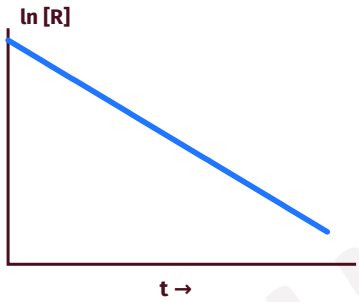
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[R] vs t



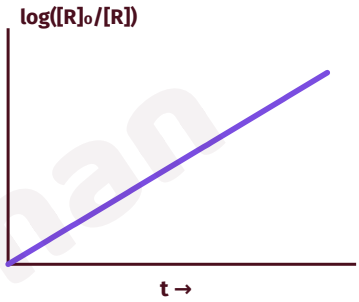
exponential: $[R] = [R]_0 e^{-kt}$

ln [R] vs t



slope = −k

log([R]₀/[R]) vs t



slope = k/2.303

First order examples (NCERT)	Rate law
Hydrogenation of ethene: $C_2H_4(g) + H_2(g) \rightarrow C_2H_6(g)$	rate = $k[C_2H_4]$
Radioactive decay: $^{226}_{88}Ra \rightarrow {}^{4}_{2}He + {}^{222}_{86}Rn$ (saare natural aur artificial decays)	rate = $k[Ra]$
$N_2O_5(g) \rightarrow 2NO_2(g) + ½O_2(g)$; $N_2O(g) \rightarrow N_2(g) + ½O_2(g)$	rate = $k[N_2O_5]$; $k[N_2O]$

⚠ Galti mat karna. log ke andar **$[R]_0/[R]$** (bada ÷ chhota) rakho, taaki log positive aaye. Ulta likhoge to k negative aa jaayega.

Ruko aur socho

For a first order reaction, which plot is a straight line?

☐ [R] vs t ☐ ln[R] vs t ☐ [R] vs 1/t

Ab tumhari baari *answers topic ke end mein*

1. A first order reaction has $k = 0.0693\ min^{-1}$. What % remains after 20 min?

Answer

2. $k = 2.303 \times 10^{-2}\ s^{-1}$. Time for [R] to become $[R]_0/10$?

Answer

Exam mein aise likho

Integrated rate equation for a first order reaction

For R → P, rate = $-d[R]/dt = k[R]$, so $d[R]/[R] = -k\,dt$. Integrating, $\ln[R] = -kt + I$. At $t = 0$, $[R] = [R]_0$, so $I = \ln[R]_0$. Hence $\ln[R] = -kt + \ln[R]_0$, i.e. $\ln([R]/[R]_0) = -kt$, or $k = (1/t)\ln([R]_0/[R]) = (2.303/t)\log([R]_0/[R])$. A plot of ln[R] vs t is a straight line with slope −k and intercept $\ln[R]_0$.

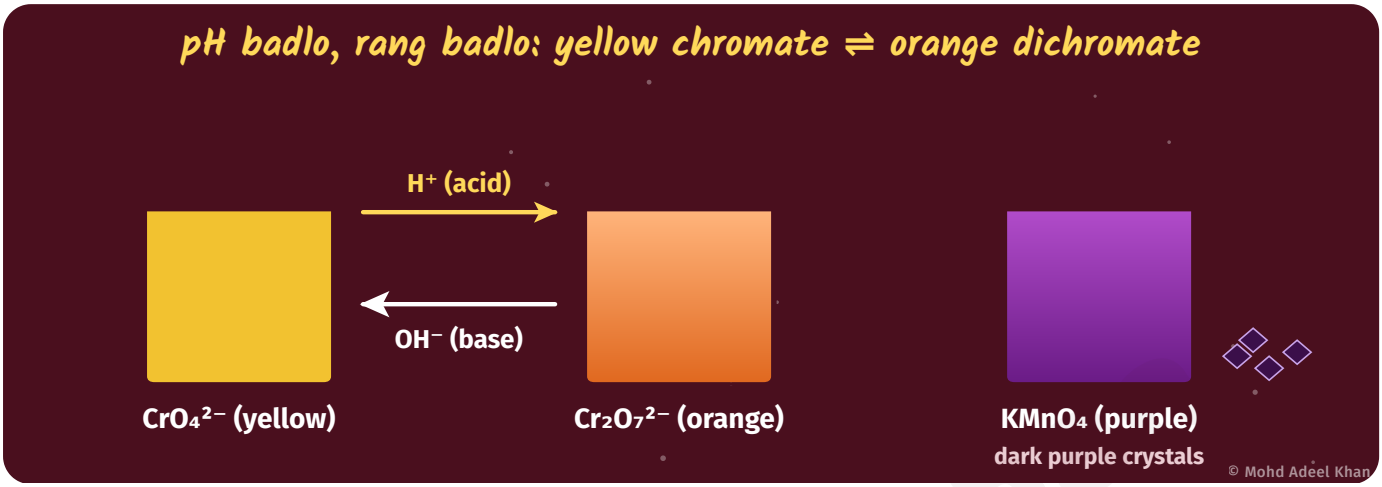
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Topic 5

Oxides, $K_2Cr_2O_7$ aur $KMnO_4$

Lab ke do sabse famous oxidising agents



Transition metals ke **oxides** oxidation state ke saath basic se acidic ho jaate hain. Inke **oxoanions** mein do bahut important hain: **potassium dichromate** (orange, $K_2Cr_2O_7$) aur **potassium permanganate** (purple, $KMnO_4$). Dono strong oxidising agents hain aur titration (volumetric analysis) mein use hote hain. Is topic mein inki **preparation**, **structure** aur **reactions** padhenge.

Is topic mein

Oxides aur oxoanions ka trend (Exercise 4.25)	31
$K_2Cr_2O_7$: preparation, structure (Exercise 4.26)	32
$K_2Cr_2O_7$ ka oxidising action (Exercises 4.14, 4.15)	33
$KMnO_4$: preparation, structure, properties (Example 4.9)	34
$KMnO_4$ ka oxidising action (Exercise 4.16)	35
Ek nazar mein + quick check	36

Padhne ke baad tick karo

- ☐ Oxides ka basic → acidic trend aur Exercise 4.25 samjha sakta hoon
- ☐ $K_2Cr_2O_7$ ki preparation aur chromate ⇌ dichromate likh sakta hoon
- ☐ $KMnO_4$ ki preparation, structure aur Example 4.9 likh sakta hoon
- ☐ Acidic aur neutral medium ki ionic equations balance kar sakta hoon

Adeel Sir

Adeel Sir kehte hain

Do numbers yaad rakho: $Cr_2O_7^{2-}$ acid mein **6 electrons** leta hai (→ $2Cr^{3+}$), aur MnO_4^- acid mein **5 electrons** (→ Mn^{2+}), neutral mein 3 (→ MnO_2). Saari equations inhi se balance hoti hain.

Kab kya hua

1659
Glauber: mineral chameleon (manganate)

1797
Vauquelin: crocoite se chromium

1850s
Condy's crystals ($KMnO_4$) disinfectant

Aaj
titration, water treatment

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Section 4.4.1

Samajh aaya? Rate karo ☆ ☆ ☆

Oxides aur oxoanions

Adeel Sir ka example

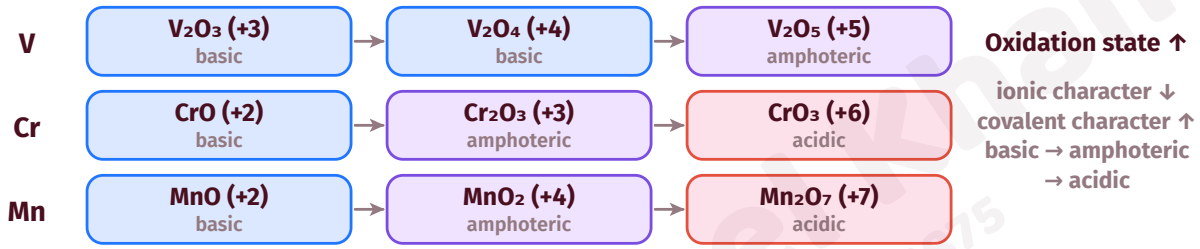
Adeel Sir ki class mein ek question aaya: 'Ek hi metal ka oxide basic bhi aur acidic bhi kaise?' Answer: Mn ka MnO acid mein ghul kar Mn^{2+} salt deta hai (basic), par Mn_2O_7 paani mein $HMnO_4$ (acid) deta hai. Farq hai **oxidation state** ka.

Oxides

High temperature par metal + O_2 se bante hain. Sc ko chhodkar saare metals **MO** oxide (ionic) banate hain. Highest oxidation number **group number** ke barabar: Sc_2O_3 se Mn_2O_7 tak. Group 7 ke baad: Fe ka Fe_2O_3 se upar koi oxide nahi.

Oxidation state badhne par

Ionic character ghatta hai: Mn_2O_7 covalent **green oil** hai; CrO_3, V_2O_5 ke melting point kam. Higher oxides **acidic**: $Mn_2O_7 \rightarrow HMnO_4$; $CrO_3 \rightarrow H_2CrO_4$, $H_2Cr_2O_7$. Oxocations: VO_2^+ (V(V)), VO^{2+} (V(IV)), TiO^{2+} (Ti(IV)).



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V_2O_5 amphoteric hai (mainly acidic): alkali ke saath VO_4^{3-} aur acid ke saath VO_2^+ salts. V_2O_4 acids mein ghul kar VO^{2+} salts deta hai. CrO basic hai, par Cr_2O_3 amphoteric.

Exam mein aise likho

Exercise 4.25: Give examples and suggest reasons: (i) the lowest oxide of a transition metal is basic, the highest is amphoteric/acidic (ii) a transition metal exhibits its highest oxidation state in oxides and fluorides (iii) the highest oxidation state is exhibited in oxoanions of a metal.

(i) MnO is basic while Mn_2O_7 is acidic; CrO is basic, Cr_2O_3 amphoteric and CrO_3 acidic. In the low oxidation state the oxide is ionic and the metal can give electrons, so it is basic. In the high oxidation state the metal (small, highly charged) forms covalent bonds and attracts electrons, so the oxide is acidic. (ii) Oxygen and fluorine are small and highly electronegative, so they can oxidise the metal to its highest oxidation state, e.g. VF_5, CrF_6, Mn_2O_7. (iii) In oxoanions such as MnO_4^- (Mn +7) and $Cr_2O_7^{2-}$ (Cr +6), oxygen forms multiple bonds with the metal and stabilises the highest oxidation state.

⚠ Galti mat karna. Mn_2O_7 ko ionic solid mat likhna: yeh **covalent green oil** hai, jisme Mn–O–Mn bridge hai.

Ruko aur socho

Which oxide is amphoteric?

☐ CrO ☐ Cr_2O_3 ☐ CrO_3

Ab tumhari baari *answers topic ke end mein*

1. Which is more basic, CrO or CrO_3?

Answer

2. Write the formula of the oxocation of vanadium(IV).

Answer

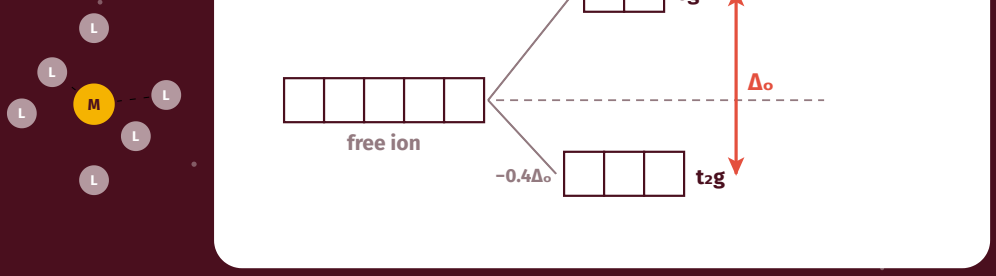
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Topic 5

Crystal field theory

d orbitals ka splitting aur high/low spin

Ligands aaye, d orbitals ki energy do hisson mein bant gayi



eg: ligands ki taraf (dx^2-y^2, dz^2)

t_{2g}: ligands ke beech (dxy, dyz, dxz)

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VBT colour aur weak/strong ligand ka farq nahi samjha paati. **Crystal field theory (CFT)** ek **electrostatic model** hai: ligands ko point charges (anions) ya point dipoles (neutral molecules) maan kar dekhti hai ki unke aane se metal ke **paanch d orbitals** ki energy kaise **split** hoti hai. Is topic mein octahedral aur tetrahedral splitting, **spectrochemical series** aur **Δ_o vs P** se high spin/low spin padhenge.

Is topic mein

Octahedral field: t_{2g} aur eg splitting (Exercise 5.16)	**31**
Spectrochemical series, Δ_o vs P, high/low spin (Intext 5.10)	**32**
Tetrahedral field; Exercise 5.17	**33**
NCERT exercises: CFSE, d occupancy (5.18, 5.23)	**34**
Ek nazar mein + quick check	**35**

Padhne ke baad tick karo

- ☐ Octahedral field mein t_{2g} aur eg splitting draw kar sakta hoon (Exercise 5.16)
- ☐ Spectrochemical series aur weak/strong field ligands bata sakta hoon (Exercise 5.17)
- ☐ Δ_o aur P se d^4 to d^7 configuration likh sakta hoon (Exercise 5.18, Intext 5.10)
- ☐ Tetrahedral splitting aur $\Delta_t = (4/9)\Delta_o$ samjha sakta hoon

Adeel Sir

Adeel Sir kehte hain

Ek line mein: ligand jitna **strong**, Δ_o utna **bada**; Δ_o **pairing energy P** se bada ho to electrons t_{2g} mein pair ho jaate hain (**low spin**), warna e_g mein chale jaate hain (**high spin**).

Kab kya hua

1929
Bethe: crystal field theory

1932
Van Vleck: magnetism par lagu

1950s
complexes ke rang samjhe gaye

Aaj
ligand field theory

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Section 5.5.4 (a)

Samajh aaya? Rate karo ☆ ☆ ☆

Crystal field theory: octahedral splitting

Adeel Sir ka example

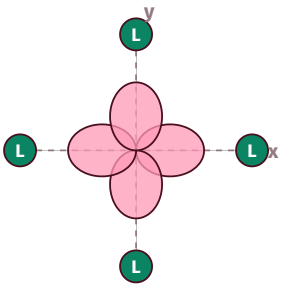
Adeel Sir ke paas paanch dost ek line mein khade hain, sab ki height barabar (degenerate d orbitals). Ab chaar taraf se kuch log dhakka dene aaye (ligands). Jo seedha unki taraf khade the, unhe zyada dhakka laga aur woh upar ho gaye; jo beech mein the, unhe kam. Yahi hai **crystal field splitting**.

CFT ki assumptions

Metal–ligand bond **ionic** maana jaata hai, sirf electrostatic attraction. Ligands = **point charges** (anions) ya **point dipoles** (neutral). Akele gaseous ion mein paanchon d orbitals **degenerate** (same energy). Spherical field mein bhi degenerate, par ligands ka field asymmetrical hota hai, isliye degeneracy toot jaati hai: **splitting**.

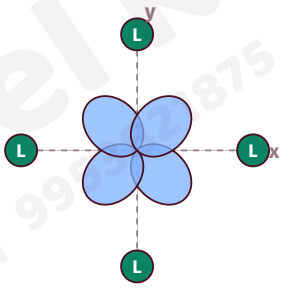
Octahedral field

Chhe ligands x, y, z axes par. $d_{x^2-y^2}$ aur d_{z^2} ke lobes **ligands ki taraf**: zyada repulsion, energy upar: $\mathbf{e_g}$ set. d_{xy}, d_{yz}, d_{xz} **axes ke beech**: kam repulsion, energy neeche: $\mathbf{t_{2g}}$ set. Dono ka gap: $\mathbf{\Delta_o}$ (crystal field splitting).



dx^2-y^2 (eg set)

lobes ligands ki taraf: zyada repulsion



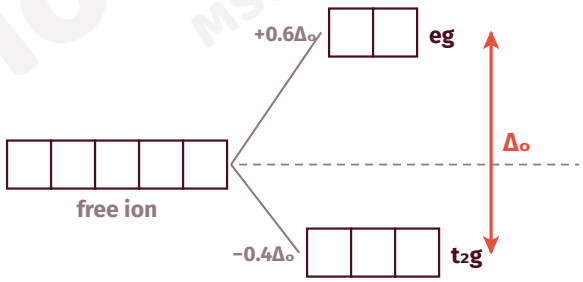
dxy (t_2g set)

lobes ligands ke beech: kam repulsion

dz^2 bhi axis par (eg);

dyz, dxz bhi axes ke beech (t_2g)

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Barycentre (average energy) same rehta hai:
$2 \times (+0.6\Delta_o) + 3 \times (-0.4\Delta_o) = 0$

eg ka har orbital 3/5 Δ_o upar,
t_{2g} ka har orbital 2/5 Δ_o neeche

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Exam mein aise likho

Exercise 5.16: Draw figure to show the splitting of d orbitals in an octahedral crystal field.

In an octahedral field the six ligands approach along the x, y and z axes. The $d_{x^2-y^2}$ and d_{z^2} orbitals point towards the ligands and experience more repulsion, so their energy is raised (e_g set, $+0.6\Delta_o$). The d_{xy}, d_{yz} and d_{xz} orbitals lie between the axes and are lowered in energy (t_{2g} set, $-0.4\Delta_o$) relative to the average energy in a spherical field. The energy gap between the two sets is the crystal field splitting energy, Δ_o.

(Draw the diagram shown above.)

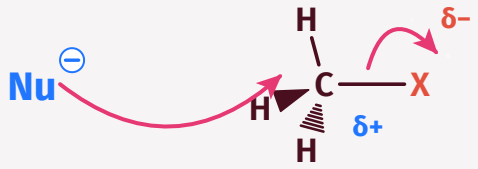
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Topic 4

Nucleophilic substitution

SN2 aur SN1: haloalkanes ki sabse important reaction

Nucleophile δ+ carbon par hamla karta hai, X^- bahar



Nu⁻ ka electron pair andar, C–X ka pair X ke saath bahar

Nucleophiles (electron-rich)

OH^-	H_2O	RO^-
I^-	CN^-	NH_3
NO_2^-	$RCOO^-$	H^-

Rate: SN2 (ek step, 2nd order) ya SN1 (do step, carbocation, 1st order)

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C–X ka carbon δ+ hai. **Nucleophile** (electron-rich species) is carbon par hamla karta hai aur halogen X^- ban kar nikal jaata hai (leaving group): yeh **nucleophilic substitution** hai. Do mechanisms: **S_N2** (ek step) aur **S_N1** (do step, carbocation ke zariye); kaunsa chalega, yeh substrate ki structure aur solvent tay karte hain.

Is topic mein

Nucleophiles aur products (Table 6.4), ambident (Example 6.5, Exercise 6.8)	27
SN2 mechanism: backside attack, inversion, steric hindrance (Example 6.6)	28
SN1 mechanism: carbocation, stability order (Example 6.7)	29
SN1 vs SN2 table, Intext 6.7, 6.8	30
NCERT Exercises 6.9, 6.15, 6.16, 6.17	31
Ek nazar mein + quick check	32

Padhne ke baad tick karo

- ☐ Nucleophile ke hisaab se product bata sakta hoon (Table 6.4)
- ☐ Ambident nucleophile: KCN vs AgCN, KNO_2 vs $AgNO_2$ samjha sakta hoon
- ☐ SN2 aur SN1 ke mechanism, rate law aur energy profile bana sakta hoon
- ☐ SN1/SN2 reactivity order nikaal sakta hoon (Examples 6.6, 6.7)

Adeel Sir

Adeel Sir kehte hain

Do questions se mechanism pehchano: **carbon kitna bheed-bhaad wala hai?** (SN2 ke liye kam bheed) aur **carbocation kitna stable banega?** (SN1 ke liye zyada stable).

Kab kya hua

1896
Walden: inversion ki khoj

1937
Hughes, Ingold: SN1, SN2 mechanisms

Aaj
drug synthesis ka aadhaar

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Section 6.7.1 (I)

Samajh aaya? Rate karo ☆ ☆ ☆

Nucleophile badlo, product badlo

$$Nu^- + R{-}X \longrightarrow R{-}Nu + X^-$$

Reagent	Nucleophile	Product R–Nu	Class
NaOH (KOH) / H_2O	HO^- / H_2O	ROH	Alcohol
NaOR′	$R'O^-$	ROR′	Ether
NaI	I^-	R–I	Alkyl iodide
NH_3 / $R'NH_2$ / $R'R''NH$	same	RNH_2 / RNHR′ / RNR′R″	1° / 2° / 3° amine
KCN	$^-$:C≡N:	RCN	Nitrile (cyanide)
AgCN	Ag–C≡N:	RNC	Isonitrile (isocyanide)
KNO_2	O=N–O^-	R–O–N=O	Alkyl nitrite
$AgNO_2$	Ag–O–N=O	$R{-}NO_2$	Nitroalkane
R′COOAg	$R'COO^-$	R′COOR	Ester
$LiAlH_4$ / R'^-M^+	H^- / R'^-	RH / RR′	Hydrocarbon / alkane

Ambident nucleophiles

Jin nucleophiles ke paas **do alag nucleophilic sites** hon aur woh kisi ek se bhi attack kar sakein. **CN^-**: C se attack → alkyl cyanide (RCN); N se attack → isocyanide (RNC). **NO_2^-**: O se attack → alkyl nitrite (R–O–N=O); N se attack → nitroalkane ($R{-}NO_2$).

Exam mein aise likho

Example 6.5: Haloalkanes react with KCN to form alkyl cyanides as main product while AgCN forms isocyanides as the chief product. Explain.

KCN is predominantly ionic and gives free cyanide ions in solution. Although both C and N can donate an electron pair, attack takes place mainly through the carbon atom, because the C–C bond formed is more stable than a C–N bond; so alkyl cyanide is the main product. AgCN is mainly covalent; the carbon is bonded to Ag, so only the lone pair on nitrogen is free to attack, and isocyanide is the chief product.

Exam mein aise likho

Exercise 6.8: What are ambident nucleophiles? Explain with an example.

Nucleophiles which have two nucleophilic sites and can attack through either of them are called ambident nucleophiles. Example: the nitrite ion, O=N–O^-. Attack through oxygen gives alkyl nitrite, R–O–N=O (e.g. with KNO_2); attack through nitrogen gives nitroalkane, $R{-}NO_2$ (e.g. with $AgNO_2$). The cyanide ion (C or N) is another example.

Ab tumhari baari *answers topic ke end mein*

1. $CH_3CH_2Br + CH_3ONa \rightarrow$?

Answer

2. CH_3CH_2Br + excess $NH_3 \rightarrow$?

Answer

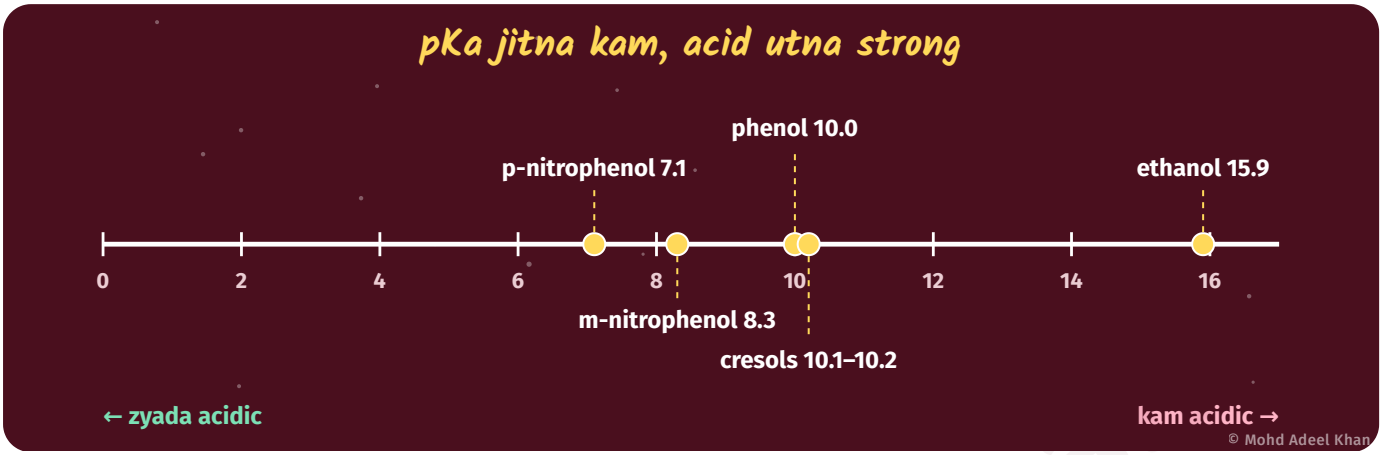
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Topic 4

Acidity aur O–H bond ki reactions

Alcohol, phenol, paani: kaun kitna acidic?



Alcohols aur phenols **active metals** (Na, K, Al) se H_2 dete hain aur proton chhod sakte hain, yaani **Brönsted acids** hain. Par phenol alcohol se kahin zyada acidic hai: woh NaOH se bhi react karta hai. Is topic mein yeh farq **resonance** aur **sp² carbon** se samjhenge, substituents ka asar (pKa values) dekhenge, aur esterification padhenge.

Is topic mein

Metals aur NaOH ke saath; alcohols ki acidity	**24**
Phenol zyada acidic kyun: phenoxide resonance, pKa (Table 7.3)	**25**
Acid strength order, nitrophenoxide resonance (Example 7.4, Intext 7.8, Exercises 7.14, 7.15)	**26**
Esterification aur acetylation: aspirin	**27**
Ek nazar mein + quick check	**28**

Padhne ke baad tick karo

- [ ] Alcohols ki acidity 1° > 2° > 3° aur paani se kam kyun, samjha sakta hoon
- [ ] Phenoxide ion ki resonance structures bana sakta hoon (Intext 7.8)
- [ ] Nitro aur alkyl groups ka phenol ki acidity par asar bata sakta hoon (Example 7.4)
- [ ] Esterification aur aspirin ka banna likh sakta hoon

Adeel Sir

Adeel Sir kehte hain

Acidity ka question hamesha **conjugate base ki stability** se solve karo: phenoxide (charge phaila hua) zyada stable, alkoxide (charge O par atka) kam stable.

Kab kya hua

1853
Gerhardt: acetylsalicylic acid

1897
Hoffmann: shuddh aspirin

1923
Brönsted–Lowry acid theory

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Section 7.4.4 (a)

Samajh aaya? Rate karo ☆ ☆ ☆

O–H bond toot-ta hai: acidity

Adeel Sir ka example

Adeel Sir ne beaker mein ethanol liya aur sodium ka chhota tukda daala: bulbule (H_2) nikle. Phenol + NaOH bhi ghul gaya. Par ethanol + NaOH mein koi khaas reaction nahi. Matlab dono acidic hain, par phenol kahin zyada.

Active metals aur NaOH ke saath

$$2R\text{–}OH + 2Na \xrightarrow[\text{sodium alkoxide}]{} 2R\text{–}ONa + H_2$$

$$6(CH_3)_3C\text{–}OH + 2Al \xrightarrow[\text{aluminium tert-butoxide}]{} 2[(CH_3)_3C\text{–}O]_3Al + 3H_2$$

$$2C_6H_5OH + 2Na \longrightarrow 2C_6H_5ONa + H_2$$

$$C_6H_5OH + NaOH \xrightarrow[\text{sodium phenoxide}]{} C_6H_5ONa + H_2O$$

Yeh reactions dikhati hain ki alcohols aur phenols **Brönsted acids** hain: strong base ko proton de sakte hain.

Alcohols ki acidity

Acidity polar O–H bond ki wajah se. Electron releasing alkyl groups (–CH_3, –C_2H_5) O par electron density badhate hain, O–H ki polarity ghatti hai: acid strength **1° > 2° > 3°**.

Paani se bhi kamzor acid

$R\text{–}O^- + H\text{–}OH \rightarrow R\text{–}OH + OH^-$: paani alkoxide ko proton de deta hai, yaani **paani zyada strong acid** hai. Isliye alkoxide ($NaOC_2H_5$) NaOH se zyada **strong base** hai.

Alcohols Brönsted base bhi

O par do **lone pairs** hain, isliye alcohol proton le bhi sakta hai: $R\text{–}OH + H^+ \rightarrow R\text{–}OH_2^+$. Yahi protonation dehydration aur HX wali reactions ka pehla step hai (Topic 5).

⚠ Galti mat karna. 'Alcohol acidic hai, isliye NaOH se react karega': galat. Alcohol paani se bhi kamzor acid hai, NaOH ke saath equilibrium bilkul peeche rehta hai. Phenol NaOH se react karta hai.

Ruko aur socho

Which is the strongest acid?

☐ ethanol ☐ water ☐ tert-butyl alcohol

Exam mein aise likho

Adeel Sir's question: Show by a reaction that alcohols are weaker acids than water. Which is the stronger base, sodium ethoxide or sodium hydroxide?

When an alkoxide is treated with water, water donates a proton to it: $C_2H_5O^- + H\text{–}OH \rightarrow C_2H_5OH + OH^-$. Water acts as the stronger acid (better proton donor) and ethoxide as the stronger base. Hence alcohols are weaker acids than water, and sodium ethoxide is a stronger base than sodium hydroxide.

Ab tumhari baari *answers topic ke end mein*

1. Write the reaction of ethanol with sodium.

Answer

2. Arrange in decreasing acid strength: 1°, 2°, 3° alcohols.

Answer

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Topic 5

α-Hydrogen, Cannizzaro aur uses

Aldol se badi molecule, Cannizzaro se alcohol aur acid



Carbonyl group ke bagal wale carbon (**α-carbon**) ka H **acidic** hota hai. Dilute alkali isko nikaal kar **enolate** banata hai jo doosre molecule ke C=O par attack karta hai: yeh **aldol condensation** hai. Jin aldehydes mein α-H nahi (HCHO, C_6H_5CHO), woh concentrated alkali mein **Cannizzaro reaction** dete hain. Aromatic aldehydes ki ring par **meta** substitution, aur aakhir mein inke **uses**.

Is topic mein

α-H ki acidity aur aldol condensation	34
Cross aldol aur Claisen–Schmidt (Exercises 8.9, 8.16 (iii), 8.17 (vii))	35
Cannizzaro, electrophilic substitution (Exercises 8.10, 8.16 (ii))	36
NCERT: aldol ya Cannizzaro? (Exercises 8.7, 8.8)	37
Uses of aldehydes and ketones	38
Ek nazar mein + quick check	39

Padhne ke baad tick karo

- ☐ α-H acidic kyun hai, samjha sakta hoon
- ☐ Aldol aur cross aldol ke products likh sakta hoon (Exercises 8.8, 8.9)
- ☐ Cannizzaro reaction pehchaan kar likh sakta hoon (Exercises 8.7, 8.10)
- ☐ Aldehydes aur ketones ke uses bata sakta hoon

Adeel Sir

Adeel Sir kehte hain

Pehla question hamesha: **α-carbon par H hai?** Haan: aldol (dilute alkali). Nahi (aldehyde): Cannizzaro (concentrated alkali).

Kab kya hua

1853 Cannizzaro reaction

1872 Wurtz: aldol

1881 Claisen–Schmidt

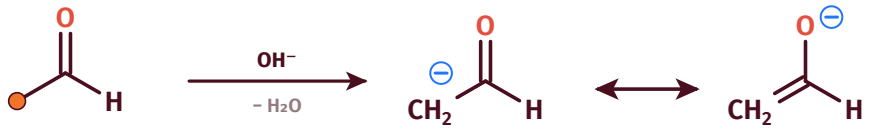
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Section 8.4.4 (i)

Samajh aaya? Rate karo ☆ ☆ ☆

α-Hydrogen aur aldol condensation

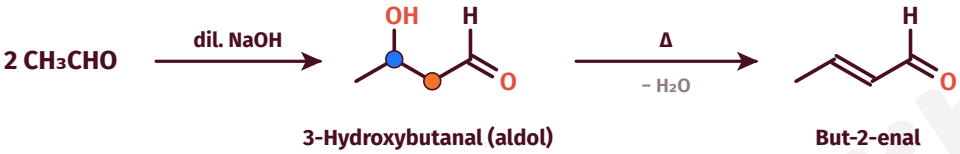


ethanal (α-H) — enolate ion: resonance se stable

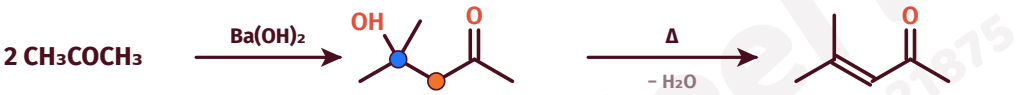
α-H acidic kyun?
1. C=O ka strong electron withdrawal
2. conjugate base (enolate) resonance se stable

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α-H ki acidity: carbonyl group ka **strong electron withdrawing** effect aur conjugate base (**enolate**) ka **resonance stabilisation**. Kam se kam **ek α-H** wale aldehydes aur ketones **dilute alkali** ke saath **β-hydroxy aldehyde (aldol)** ya **β-hydroxy ketone (ketol)** dete hain. Aldol aur ketol aasaani se **paani kho kar α,β-unsaturated** carbonyl compound dete hain: **aldol condensation**.



3-Hydroxybutanal (aldol) — But-2-enal



4-Hydroxy-4-methylpentan-2-one (ketol) — 4-Methylpent-3-en-2-one

α-carbon
β-carbon (–OH)

Aldol = ald + ol: β-hydroxy aldehyde. Ketone se ketol. Garam karne par α-H aur β-OH paani ban kar nikalte: α,β-unsaturated product.

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$2CH_3CHO \xrightarrow{\text{dil. NaOH}} CH_3CH(OH)CH_2CHO \xrightarrow[-H_2O]{\Delta} CH_3CH{=}CHCHO$

Exam mein aise likho

Adeel Sir's question: What is aldol condensation? Explain with ethanal.

Aldehydes and ketones having at least one α-hydrogen react in the presence of dilute alkali to form β-hydroxy aldehydes (aldols) or β-hydroxy ketones (ketols). These readily lose water on heating to give α,β-unsaturated carbonyl compounds. This reaction is called aldol condensation. Example: $2CH_3CHO \rightarrow$ (dil. NaOH) $CH_3CH(OH)CH_2CHO$ (3-hydroxybutanal) $\rightarrow$ (heat, $-H_2O$) $CH_3CH{=}CHCHO$ (but-2-enal).

★ JEE/NEET extra

Mechanism: OH^- α-H nikaal kar **enolate** banata hai; enolate (nucleophile) doosre molecule ke **C=O carbon** par attack karta hai; alkoxide paani se H^+ le kar aldol banta hai.

Ruko aur socho

Which does NOT undergo aldol condensation?

☐ CH_3CHO ☐ CH_3COCH_3 ☐ C_6H_5CHO

Ab tumhari baari *answers topic ke end mein*

1. Aldol product of propanal (dil. NaOH)?

Answer

2. Dehydrated product of the propanal aldol?

Answer

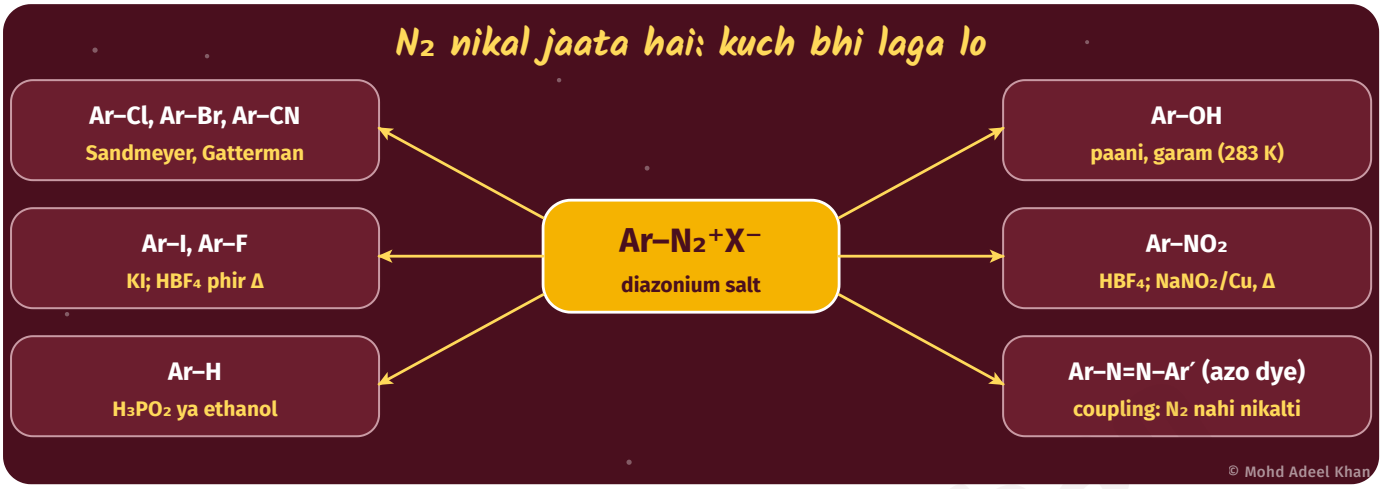
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Topic 6

Diazonium salts

Aromatic chemistry ka sabse useful bridge



Diazonium salts ka general formula **$RN_2^+X^-$** hai (X^- = Cl^-, Br^-, HSO_4^-, BF_4^-). Aliphatic 1° amines se bane alkyldiazonium salts bahut unstable hain; **aromatic** 1° amines se bane arenediazonium salts **thande (273–278 K)** solution mein thodi der stable. N_2 bahut achha leaving group hai, isliye diazonium group ki jagah Cl, Br, I, F, CN, OH, H, NO_2 laga sakte hain; aur coupling se rangeen **azo dyes**.

Is topic mein

Diazotisation, stability, physical properties (Exercises 9.3 (vi), 9.7 (ii))	**34**
N2 ka displacement: Sandmeyer, Gatterman aur doosri (Exercise 9.11)	**35**
Coupling reactions aur importance (Exercise 9.7 (iv))	**36**
Example 9.5, Intext 9.9	**37**
Ek nazar mein + quick check	**38**

Padhne ke baad tick karo

- ☐ Diazotisation aur aryl diazonium salts ki stability samjha sakta hoon (Exercise 9.3 (vi))
- ☐ Sandmeyer, Gatterman aur doosri N_2-displacement reactions likh sakta hoon
- ☐ Coupling reaction se azo dyes likh sakta hoon (Exercise 9.7 (iv))
- ☐ Diazonium route se conversions kar sakta hoon (Example 9.5, Intext 9.9)

Adeel Sir

Adeel Sir kehte hain

Ring par jo group direct nahi lag sakta (F, I, CN, OH, ya H se $-NH_2$ hatana), woh **$-NH_2 \rightarrow -N_2^+ \rightarrow$ group** raaste se lagta hai.

Kab kya hua

1858 Griess: diazonium salts

1884 Sandmeyer reaction

1890 Gatterman reaction

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Section 9.7, 9.8

Samajh aaya? Rate karo ☆ ☆ ☆

Diazonium salts: banana aur properties

Naam

Parent hydrocarbon ke naam ke baad **'diazonium'** aur anion ka naam: $C_6H_5N_2^+Cl^-$ = **benzenediazonium chloride**; $C_6H_5N_2^+HSO_4^-$ = benzenediazonium hydrogensulphate.

Stability

Arenediazonium ion ka **positive charge resonance se benzene ring par phail** jaata hai, isliye yeh alkyldiazonium se zyada stable. Phir bhi sirf thande mein, thodi der ke liye.

Diazotisation (NCERT 9.7)

Aniline + nitrous acid (**$NaNO_2$ + HCl** se bana) **273–278 K** par: benzenediazonium chloride. Diazonium salts aam taur par **store nahi** karte; banate hi turant use.

$$C_6H_5NH_2 + NaNO_2 + 2HCl \xrightarrow{273\text{–}278\ K} C_6H_5N_2^+Cl^- + NaCl + 2H_2O$$

Physical properties (NCERT 9.8)

Benzenediazonium chloride: **colourless crystalline solid**, paani mein aasaani se soluble; thande mein stable par **garam karne par paani se react**. **Sookhi (dry) state** mein aasaani se decompose. **Benzenediazonium fluoroborate** paani mein insoluble aur **room temperature par stable**.

Exam mein aise likho

Exercises 9.3 (vi), 9.7 (ii): (a) Diazonium salts of aromatic amines are more stable than those of aliphatic amines. (b) Write a short note on diazotisation.

(a) In arenediazonium ions, the positive charge is dispersed over the benzene ring by resonance, which stabilises them; they are stable for a short time in solution at low temperature (273–278 K). Alkyldiazonium ions have no such resonance stabilisation, so they decompose immediately, giving nitrogen gas. (b) Diazotisation: the conversion of a primary aromatic amine into its diazonium salt by treatment with nitrous acid ($NaNO_2$ + HCl) at 273–278 K: $C_6H_5NH_2 + NaNO_2 + 2HCl \rightarrow C_6H_5N_2^+Cl^- + NaCl + 2H_2O$. The diazonium salt is used immediately because it is unstable.

⚠ Galti mat karna. Diazotisation **thande mein (273–278 K)** hi karna. Garam karne par diazonium salt paani se react karke **phenol** de deta hai.

Ruko aur socho

Which diazonium salt is stable at room temperature?

- ☐ benzenediazonium chloride
- ☐ benzenediazonium fluoroborate
- ☐ methyldiazonium chloride

★ JEE/NEET extra

Sookhe (dry) diazonium salts jhatke ya garmi se **explosively** decompose ho sakte hain. Isliye inhe solution mein bana kar turant use karte hain.

Ab tumhari baari *answers topic ke end mein*

1. Name $C_6H_5N_2^+HSO_4^-$.

Answer

2. Why is diazotisation done at 273–278 K?

Answer

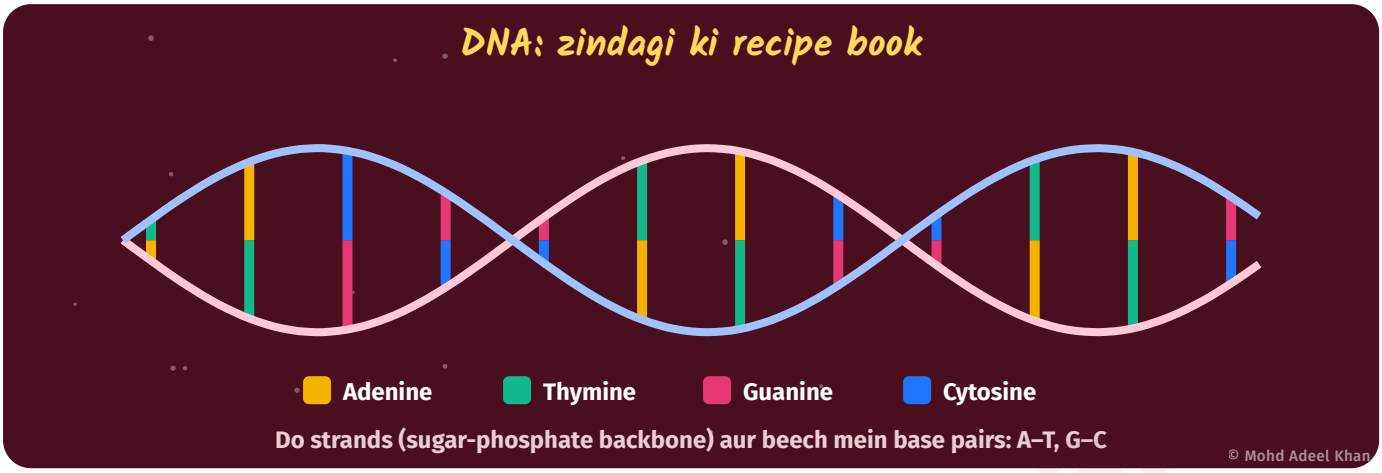
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Topic 7

Nucleic acids

Heredity ki chemistry: DNA aur RNA



Har species apni pehchaan generation dar generation bachaaye rakhti hai, aur bachchon mein maa-baap ke kuch guna (traits) aate hain. Yeh kaam **chromosomes** mein maujood **nucleic acids** karte hain. Is topic mein nucleic acids ki **chemical composition**, **nucleoside aur nucleotide**, **DNA ki double helix**, **RNA ke types** aur inke **kaam** padhenge.

Is topic mein

Chemical composition: sugar aur bases	**40**
Nucleoside, nucleotide, phosphodiester (Exercise 10.22, Intext 10.7)	**41**
DNA double helix aur RNA (Exercise 10.23, 10.25, Intext 10.8)	**42**
Kaam aur DNA vs RNA (Exercise 10.21, 10.24)	**43**
Ek nazar mein + quick check	**44**

Padhne ke baad tick karo

- [ ] DNA aur RNA ke sugar aur bases bata sakta hoon
- [ ] Nucleoside aur nucleotide ka fark likh sakta hoon (Exercise 10.22, Intext 10.7)
- [ ] DNA ke complementary strands aur RNA ke types samjha sakta hoon (Exercise 10.23, 10.25, Intext 10.8)
- [ ] Nucleic acids ke kaam aur DNA vs RNA likh sakta hoon (Exercise 10.21, 10.24)

Adeel Sir

Adeel Sir kehte hain

Nucleic acid bhi ek **polymer** hai: monomer = **nucleotide** (base + sugar + phosphate). Starch ke liye glucose, protein ke liye amino acid, waise hi DNA ke liye nucleotide.

Kab kya hua

1869
Miescher: nucleus se ‘nuclein’

1950
Chargaff: A = T, G = C

1953
Watson, Crick: double helix

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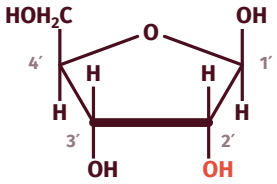
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Section 10.5, 10.5.1

Samajh aaya? Rate karo ☆ ☆ ☆

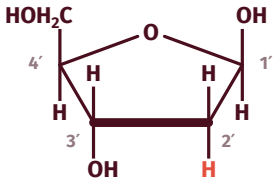
Nucleic acids kis cheez se bane hain?

Nucleic acids nucleotides ki lambi chains (**polynucleotides**) hain. Do types: **DNA** (deoxyribonucleic acid) aur **RNA** (ribonucleic acid). Poora hydrolysis karne par teen cheezein milti hain: **pentose sugar**, **phosphoric acid** aur nitrogen wale **heterocyclic bases**.



β-D-Ribose

RNA mein



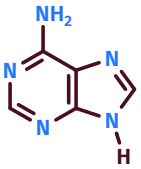
β-D-2-Deoxyribose

DNA mein: C-2′ par OH nahi

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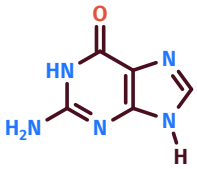
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Purines (do rings)



Adenine (A)

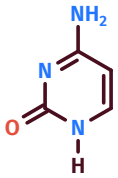
DNA + RNA



Guanine (G)

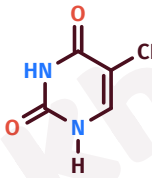
DNA + RNA

Pyrimidines (ek ring)



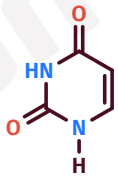
Cytosine (C)

DNA + RNA



Thymine (T)

sirf DNA



Uracil (U)

sirf RNA

Component	DNA	RNA
Sugar	β-D-2-deoxyribose	β-D-ribose
Bases	A, G, C, **T**	A, G, C, **U**
Phosphate	phosphoric acid	phosphoric acid

⚠ **Galti mat karna.** **Thymine sirf DNA** mein, **uracil sirf RNA** mein. Baaki teen bases (A, G, C) dono mein common hain.

Ruko aur socho

The sugar present in DNA is:

☐ β-D-ribose ☐ β-D-2-deoxyribose

☐ α-D-glucose

★ **JEE/NEET extra**

Chargaff's rule: DNA mein A ki quantity = T ki quantity, aur G = C. Isliye purines (A + G) = pyrimidines (T + C). A–T ke beech 2 aur G–C ke beech 3 H-bonds.

Ab tumhari baari *answers topic ke end mein*

1. Name the three products of complete hydrolysis of a nucleic acid.

Answer

2. Which base is present only in RNA?

Answer

3. Name the two purine bases.

Answer

4. Which sugar is present in RNA?

Answer

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Revision

Samajh aaya? Rate karo ☆ ☆ ☆

Fact sheet: poora chapter ek page par

1 Werner, terms

$CoCl_3 \cdot 6NH_3$: 3 AgCl (yellow, $[Co(NH_3)_6]Cl_3$)

Mohr's salt double salt; $K_4[Fe(CN)_6]$ complex

$EDTA^{4-}$ hexadentate; NO_2^-, SCN^- ambidentate

2 Nomenclature

Fe → ferrate, Cu → cuprate, Ag → argentate

bis, tris: jab ligand ke naam mein number ho

$Fe_4[Fe(CN)_6]_3$: iron(III) hexacyanidoferrate(II)

3 Isomerism

$[Co(NH_3)_3Cl_3]$: fac aur mer

$[Co(en)_3]^{3+}$: d aur l

$[Pt(NH_3)(Br)(Cl)(py)]$: 3 geometrical

4 VBT

$[Co(NH_3)_6]^{3+}$: d^2sp^3, diamagnetic

$[CoF_6]^{3-}$: sp^3d^2, 4 unpaired

$[NiCl_4]^{2-}$ sp^3; $[Ni(CN)_4]^{2-}$ dsp^2

5 CFT

d^4: $\Delta_o < P \rightarrow t_{2g}^3 e_g^1$; $\Delta_o > P \rightarrow t_{2g}^4$

$[Mn(H_2O)_6]^{2+}$: 5; $[Mn(CN)_6]^{4-}$: 1 unpaired

Tetrahedral: e neeche, t_2 upar

6 Colour

$[Ti(H_2O)_6]^{3+}$: 498 nm absorb, violet

Anhydrous $CuSO_4$ white, hydrated blue

Ruby aur emerald: Cr^{3+}

7 Carbonyls, uses

$Mn_2(CO)_{10}$: Mn–Mn; $Co_2(CO)_8$: bridging CO

Haemoglobin Fe, chlorophyll Mg, B_{12} Co

cis-platin; EDTA (lead); Wilkinson (Rh)

Ligand naam ek nazar mein

Cl^- chlorido, CN^- cyanido, OH^- hydroxido

H_2O aqua, NH_3 ammine, CO carbonyl

en ethane-1,2-diamine, py pyridine

Values jo exam mein kaam aati hain

1.73 → 5.92 μ (BM) for n = 1 to 5	**+0.6 / −0.4** e_g / t_{2g} (× Δ_o)	**4/9** $\Delta_t = (4/9)\,\Delta_o$	**498 nm** $[Ti(H_2O)_6]^{3+}$ absorbs
6 donor atoms in $EDTA^{4-}$	**3** ions from $[Co(NH_3)_5Cl]Cl_2$	**dsp^2** square planar (3d)	**d^2sp^3** inner orbital (3d)

Isomers ginne ke shortcuts

2 $[M(AA)_3]$: d, l	**3** $[MA_2(AA)_2]$: trans + cis (d, l)	**2** MA_3B_3: fac, mer	**3** square planar MABCD

NCERT padhte waqt dhyaan do

Example 5.3 (e): $Hg[Co(SCN)_4]$ ka NCERT answer Hg(I)/Co(III) hai; asal compound Hg(II)/Co(II) wala hai. Board mein NCERT answer likho.

Exercise 5.14: excess CN^- pehle Cu^{2+} ko Cu^+ banata hai; entity $[Cu(CN)_4]^{3-}$.

Exercise 5.20: $[Ni(CN)_4]^{2-}$ ka absorption visible ke bahar hai (bada splitting), isliye colourless.

Adeel Sir

Adeel Sir ka tip

Is chapter ka mantra: **oxidation state → d^n → ligand weak/strong → configuration → shape → μ.** Yeh chain yaad to aadhe questions wahin solve.

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Revision

Samajh aaya? Rate karo ☆ ☆ ☆

Board pattern test: Section A

Adeel Sir ke test rules

Book band karo. Poore test ke liye **90 minutes**, total **40 marks** (CBSE pattern). Section A: har question **1 mark**, sirf ek option.

Section A: ____ / 14

Q1 Secondary valence of Co in $[Co(NH_3)_5Cl]Cl_2$ is:
- ◯ 2
- ◯ 3
- ◯ 5
- ◯ 6

Q2 Number of ions given by $[Pt(NH_3)_4]Cl_2$ in solution:
- ◯ 2
- ◯ 3
- ◯ 4
- ◯ 5

Q3 IUPAC name of $K_3[Fe(CN)_6]$ is:
- ◯ potassium hexacyanoiron(III)
- ◯ potassium hexacyanidoferrate(III)
- ◯ potassium hexacyanidoferrate(II)
- ◯ hexacyanidoferrate(III) potassium

Q4 Which of the following shows optical isomerism?
- ◯ $[Co(NH_3)_6]^{3+}$
- ◯ trans-$[CoCl_2(en)_2]^+$
- ◯ cis-$[CoCl_2(en)_2]^+$
- ◯ $[Pt(NH_3)_2Cl_2]$

Q5 Hybridisation of $[Ni(CN)_4]^{2-}$ is:
- ◯ sp^3
- ◯ dsp^2
- ◯ d^2sp^3
- ◯ sp^3d^2

Q6 Spin only magnetic moment of $[Fe(CN)_6]^{3-}$ is:
- ◯ 0
- ◯ 1.73 BM
- ◯ 3.87 BM
- ◯ 5.92 BM

Q7 The strongest field ligand among these is:
- ◯ Cl^-
- ◯ H_2O
- ◯ NH_3
- ◯ CO

Q8 For a low spin d^6 octahedral complex the configuration is:
- ◯ $t_{2g}^4e_g^2$
- ◯ $t_{2g}^6e_g^0$
- ◯ $t_{2g}^3e_g^3$
- ◯ $t_{2g}^5e_g^1$

Q9 $[Ti(H_2O)_6]^{3+}$ appears violet because it absorbs:
- ◯ violet light
- ◯ blue-green light
- ◯ red light
- ◯ UV light

Q10 The shape of $[Fe(CO)_5]$ is:
- ◯ square pyramidal
- ◯ trigonal bipyramidal
- ◯ octahedral
- ◯ tetrahedral

Assertion-Reason *(Q11 to Q14, 1 mark each)*

(a) Both A and R are true and R is the correct explanation of A. **(b)** Both A and R are true but R is not the correct explanation of A. **(c)** A is true but R is false. **(d)** A is false but R is true.

Q11 **Assertion (A):** $[Co(en)_3]^{3+}$ is more stable than $[Co(NH_3)_6]^{3+}$.
Reason (R): en is a chelating ligand which forms rings with the metal ion.
◯ (a) ◯ (b) ◯ (c) ◯ (d)

Q12 **Assertion (A):** $[Ni(CN)_4]^{2-}$ is diamagnetic.
Reason (R): Nickel belongs to the 3d transition series.
◯ (a) ◯ (b) ◯ (c) ◯ (d)

Q13 **Assertion (A):** Tetrahedral complexes are generally high spin.
Reason (R): The crystal field splitting Δ_t is larger than Δ_o.
◯ (a) ◯ (b) ◯ (c) ◯ (d)

Q14 **Assertion (A):** Anionic ligands produce the largest crystal field splitting.
Reason (R): In CFT, ligands are treated as point charges or point dipoles.
◯ (a) ◯ (b) ◯ (c) ◯ (d)

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Poori book chahiye?

Is sample mein har chapter ke sirf do pages the. Poori **Chemophile Part 2 (for 12th Class)** mein 10 chapters ke **575 pages** hain, har topic isi style mein.

Adeel Sir

Copies aur jaankari ke liye

Mohd Adeel Khan

MSc BEd (JMI)

Ph. 9953821875

Students ke liye

- ✓ Concept pehle, ratna baad mein
- ✓ NCERT ka har question solved
- ✓ Board ke liye English answers
- ✓ Har chapter ke end mein 40 marks ka test

Teachers aur schools ke liye

- ✓ NCERT section numbers ke saath
- ✓ Har topic ke ready-made visuals
- ✓ Recap, quick check aur answer key
- ✓ Revision: fact ya formula sheet, top mistakes

Is sample mein kaun se topics the

Chapter	Sample topic	Poora chapter
1. Solutions	Topic 8: Osmosis aur osmotic pressure	66 pages
2. Electrochemistry	Topic 3: Nernst equation	59 pages
3. Chemical Kinetics	Topic 5: First order reactions aur half-life	53 pages
4. The d- and f-Block Elements	Topic 5: Potassium dichromate aur permanganate	58 pages
5. Coordination Compounds	Topic 5: Crystal field theory	55 pages
6. Haloalkanes and Haloarenes	Topic 4: Nucleophilic substitution: SN2 aur SN1	60 pages
7. Alcohols, Phenols and Ethers	Topic 4: Acidity aur O–H bond ki reactions	55 pages
8. Aldehydes, Ketones and Carboxylic Acids	Topic 5: α-Hydrogen, Cannizzaro aur uses	64 pages
9. Amines	Topic 6: Diazonium salts	52 pages
10. Biomolecules	Topic 7: Nucleic acids	53 pages

Chemophile Part 1 (for 11th Class) bhi ready hai

9 chapters, 651 pages: usi style mein, usi author ki taraf se.

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MOHD ADEEL KHAN MSc BEd (JMI) PH 9953821875 CHEMOPHILE PART 2 (FOR 12TH CLASS) ORIGINAL COPY

Chemophile

Part 2 (for 12th Class)

Free sample

Chemistry ratni nahi, samajhni hai.


Topic 5

α-Hydrogen, Cannizzaro aur uses

Aldol se badi molecule, Cannizzaro se alcohol aur acid

α-H hai ya nahi? Isi se raasta tay

Aldehyde / ketone

α-carbon par H?

hain

nahi

Aldol (dil. NaOH)

2CH₃CHO → 3-hydroxybutanal

garam karne par: but-2-enal

Cannizzaro (conc. alkali)

HCHO, C₆H₅CHO, (CH₃)₃CCHO

→ alcohol + acid ka salt

Carbonyl group ke bagal wale carbon (**α-carbon**) ka H **acidic** hota hai. Dilute alkali isko nikaal kar **enolate** banata hai jo doosre molecule ke C=O par attack karta hai: yeh **aldol condensation** hai. Jin aldehydes mein α-H nahi (HCHO, C_6H_5CHO), woh concentrated alkali mein **Cannizzaro reaction** dete hain. Aromatic aldehydes ki ring par **meta** substitution, aur aakhir mein inke **uses**.

Is topic mein

α-H ki acidity aur aldol condensation 34

Cross aldol aur Claisen–Schmidt (Exercises 8.9, 8.16 (iii), 8.17 (viii)) 35

Cannizzaro, electrophilic substitution (Exercises 8.10, 8.16 (ii)) 36

NCERT: aldol ya Cannizzaro? (Exercises 8.7, 8.8) 37

Uses of aldehydes and ketones 38

Ek nazar mein + quick check 39

Padhne ke baad tick karo

- ☐ α-H acidic kyun hai, samjha sakta hoon
- ☐ Aldol aur cross aldol ke products likh sakta hoon (Exercises 8.8, 8.9)
- ☐ Cannizzaro reaction pehchaan kar likh sakta hoon (Exercises 8.7, 8.10)
- ☐ Aldehydes aur ketones ke uses bata sakta hoon

Adeel Sir kehte hain

Pehla question hamesha: **α-carbon par H hai?** Haan: aldol (dilute alkali). Nahi (aldehyde): Cannizzaro (concentrated alkali).

Kab kya hua

1853 Cannizzaro reaction

1872 Wurtz: aldol

1881 Claisen-Schmidt

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Topic 5: Aldol, Cannizzaro

33

10	575	313	990+
chapters	colourful pages	figures aur diagrams	practice questions

Poori book ke liye contact karein: **Mohd Adeel Khan**, MSc BEd (JMI)

Ph. **9953821875**

Author

Mohd Adeel Khan

MSc BEd (JMI)

Ph. **9953821875**

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